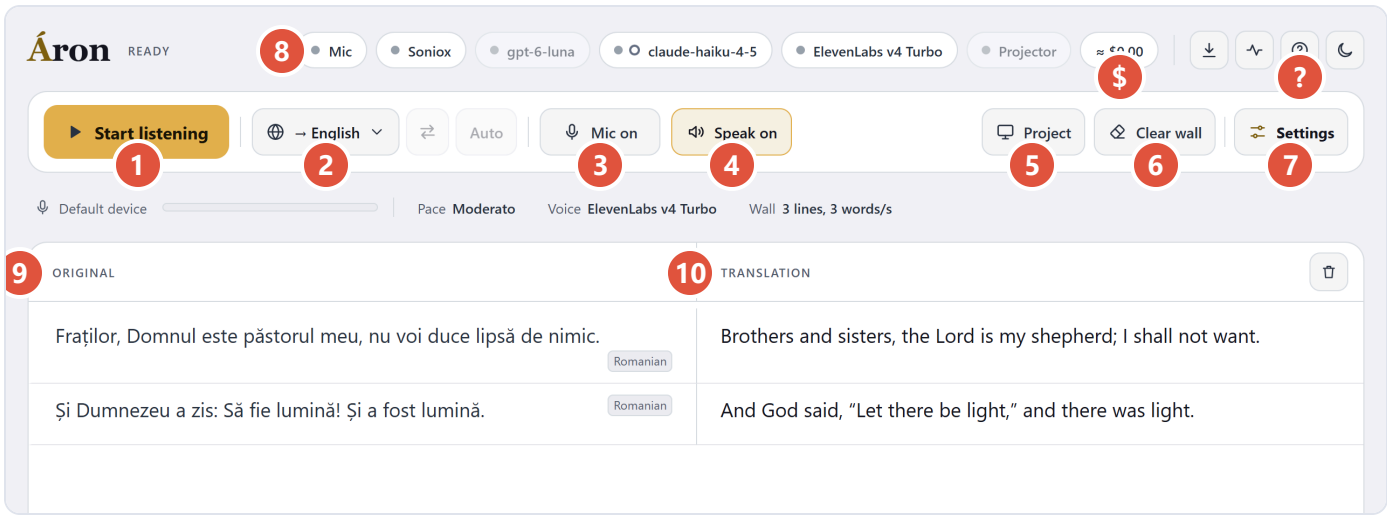
Áron — quick start guide

Live captions and spoken translation for a church service. Someone speaks; a moment later the translation appears on the screen and, if you like, is spoken aloud in another language.

Microphone the speaker → **1 · Speech-to-text** writes down what was said → **2 · Translation** into the language you pick → **Screen** captions on the projector + **3 · Voice** spoken, optional

Áron runs on an ordinary computer. It does no heavy lifting itself: it sends the sound to cloud services that recognise speech, translate and speak, and you pay each service for what you use — typically a few cents to a few dollars for a service. Each service is unlocked by an **API key**: a long password the service gives you when you open an account, which you paste into Áron once.



1 Start / stop listening · 2 the language to translate into · 3 Mic: mute it while the band plays · 4 Speak the translation aloud · 5 open the projector window · 6 clear the projector · 7 **Settings** for everything else · 8 one light for each part of the pipeline: grey is idle, green working, orange worth a look, red broken (hold the mouse over one for details) · 9 what was said · 10 the translation · $ what this run has cost so far · ? **Help** opens this guide

What you need

- A Windows computer with an internet connection (Mac and Linux work too, from the command line).
- A microphone, or better, a feed from the sound desk plugged into the computer.
- A **Soniox** account. That one key does all three stages — speech-to-text, translation, and a spoken voice. Everything else in this guide is optional and comes later.

> **The minimal setup is one key.** With just a Soniox key you get live captions (Soniox's own translation), and spoken translation if you want it — nothing else to sign up for. Parts 5 and 6 add better translation and fewer mistakes, but none of them are needed to start.

In this guide

1. Set up (about 10 minutes)
2. Your first service
3. Captions on the big screen
4. Spoken translation
5. Better translation: add an AI model
6. Fewer mistakes: the Decision Router and extra ears
7. Fine-tuning
8. What it costs
9. Keeping the service
10. Troubleshooting and shortcuts

1 Set up (about 10 minutes)

Get a Soniox key

1. Go to **console.soniox.com** and create an account.
2. Add a payment method (speech-to-text costs about **$0.12 an hour**).
3. Create an **API key** and copy it. Keep it private, like a password.

Start Áron for the first time

Most people get Áron as a single program file, **church-translate.exe**. It needs nothing installed.

1. Make a folder for Áron, for example **Documents\Aron**, and put **church-translate.exe** in it.
2. Double-click **church-translate.exe**. (Windows may warn the first time, naming the publisher: choose **More info** → **Run anyway**.) A black console window opens.
3. The first time, Áron makes its settings file, `.env`, in the same folder and opens it in **Notepad**. Find the line `SONIOX_API_KEY=` and paste your key straight after the `=` with `Ctrl`+`V`.
4. Save with `Ctrl`+`S` and close Notepad. Áron carries on by itself, and your web browser opens at `http://localhost:5000`: that page *is* Áron.

Your keys: the `.env` file

`.env` (say "dot env") is a plain text file that holds your keys. Áron reads it every time it starts. Each key is unique to your account — it is how the service knows who to bill — so treat the file like a password: keep it on the church's computer, and never email or post it. Each line is a name, an equals sign, and the key:



1. **The name**, already in the file Áron made. Leave it exactly as it is.
2. **Your key**, pasted straight after the `=` with `Ctrl`+`V`. No spaces and no quotes around it. (These keys are made up.)
3. Nothing after the `=`? That service is simply switched off. Only the Soniox line is needed; the others come in Part 5.

To change a key later, or add one from Part 5, right-click `.env` → **Open with** → **Notepad**, edit the line, save, and restart Áron. If you start Áron with the Soniox line still empty, it opens the file for you again.

> Keep the black console window open while you use Áron — closing it stops the app. Your keys stay in `.env` on this computer and are sent only to their own service.

Running from the source folder instead? (for developers, or a copy of the whole Áron folder rather than the .exe) Install **Python 3.12** from python.org with "Add python.exe to PATH" ticked, then double-click **Start Church Translate.bat**. The first time, it asks for your keys and writes `.env` for you, and spends about a minute installing what it needs.

2 Your first service

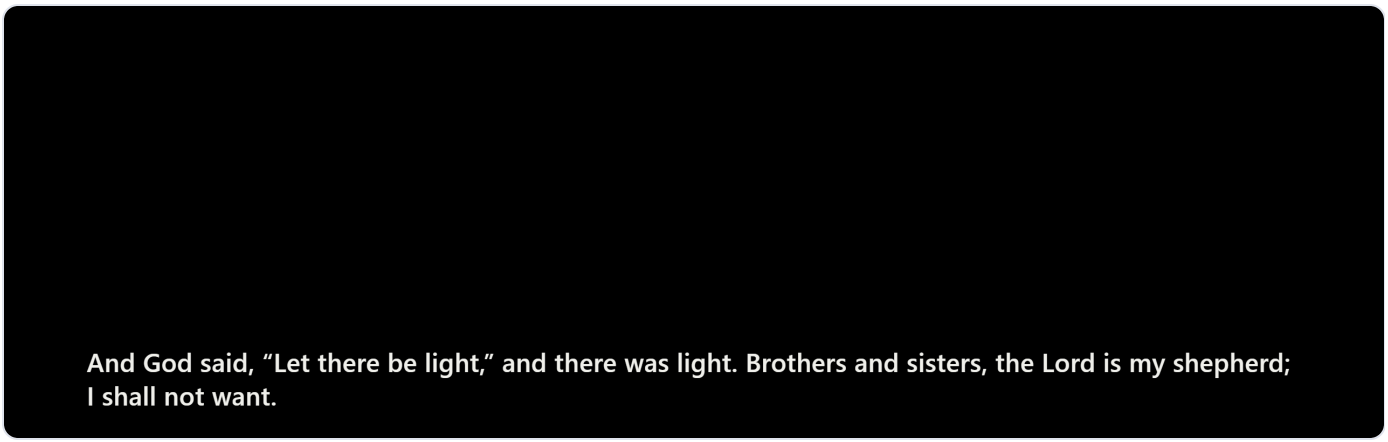
1. **Pick the language to translate into.** Click the language button (2 above, it reads → **English**) and tick one language. Áron recognises whatever language is spoken and translates it into this one. Tick **two** languages for a bilingual service: each sentence is translated into whichever of the two is *not* being spoken.
2. **Check the microphone.** Click **Settings** (7) and, under **1 Listen**, choose your microphone or sound-desk feed. When someone speaks, the small level bar under the buttons should move well into the middle; if it barely moves, raise **Gain**. Close Settings with `Esc` or a click outside it.
3. **Press Start listening** (or the `Space` bar) and speak. The **Original** column fills with what was said, and a moment later the **Translation** column fills with the translation.
4. **Press Stop to finish.** If nobody speaks for three hours, Áron stops by itself so it cannot run up a bill overnight.

That is the core pipeline — speech in, translation out. To also hear it spoken aloud, go to **Part 4**; for noticeably better translation, **Part 5**.

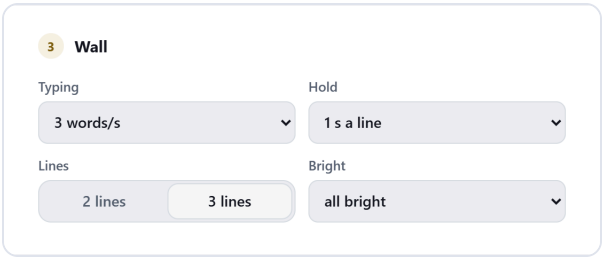
> With just the Soniox key, Áron uses Soniox's own translation (shown as **none, fast** in the Model menu, under Settings → 2 Translate). It is quick and fine for plain preaching, but it translates idioms word for word. Part 5 shows how to get much better translations.

3 Captions on the big screen

1. Click **Project** (5). A plain black caption window opens, and the **Projector** light turns green.
2. Drag it onto the projector or TV, click inside it, and press `F11` to fill the screen.
3. New translations type themselves onto the bottom of the screen as they arrive, and older lines scroll up.



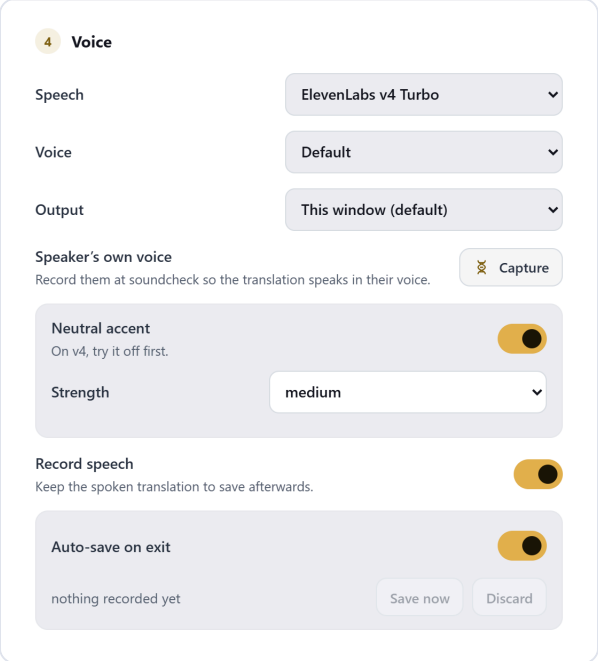
The projector window. It only ever shows the translation.



Clear wall (6, or `W`) wipes what the room is reading — for a wrong line, or between parts of the service — without touching your transcript. The bin above the Translation column empties the two columns in the main window. Under **Settings** → **3 Wall** you set how fast the words type out, how long each line holds, and whether the screen keeps 2 or 3 lines (2 are a little larger).

Also possible, set up once in `.env`: captions on your church website for phones (`CT_PUBLISH_URL`), or inside ProPresenter (`CT_PROPRESENTER_URL`). The comments in `.env.example` walk through both; once set up, each gets a switch under **Settings → 3 Wall**, and the website a **Web** light that turns red if captions stop reaching it.

4 Spoken translation



Press **Speak** (4): every translation is also read aloud, a second or two after it appears (the captions never wait for it). With your Soniox key this works straight away — no extra account to open. The voice is chosen under **Settings → 4 Voice**.

- **Speech** picks the voice service. **Soniox v2** needs no extra key (it rides on the key you already have) and costs about $0.70 an hour of speech. **ElevenLabs v4** and **v4 Turbo** have more natural voices; they need their own key.
- **Output** sends the voice to a particular device — for example a USB sound card feeding an assistive-listening transmitter, so listeners hear the translation on headphones.
- **Capture**, beside **Speaker's own voice**, records the speaker for 30 seconds at soundcheck, so the translation is spoken in their own voice. Only with their consent.

5 Better translation: add an AI model

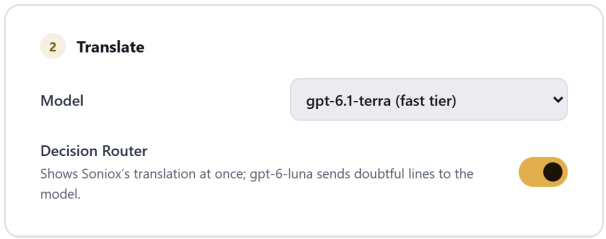
A large language model (an “LLM”, like Claude or ChatGPT) translates far better than Soniox's built-in translator. It knows idioms, regional expressions and Scripture: on a test of 21 Romanian idioms, Soniox got 2 right and Claude Haiku 16. Any one provider is enough.

Provider	Get a key at	Line to add to `.env`	Models it adds
Anthropic	console.anthropic.com	`ANTHROPIC_API_KEY=…`	claude-haiku-4-5 (the default), claude-haiku-5-5, claude-sonnet-5-5
OpenAI	platform.openai.com	`OPENAI_API_KEY=…`	gpt-6-sol, gpt-6.1-terra, gpt-6-luna — and unlocks the Decision Router (Part 6)
Google	aistudio.google.com	`GEMINI_API_KEY=…`	gemini-3.8-flash, gemini-3.5-flash-lite
xAI	console.x.ai	`XAI_API_KEY=…`	Grok models

An Anthropic API key is billed separately from a Claude.ai subscription; the same goes for OpenAI and ChatGPT.

Add the key

1. Close the black console window to stop Áron.
2. In the Áron folder, open `.env` with Notepad. Add the line from the table with your key after the `=`, and save.
3. Start Áron again. The provider's models are now available in the **Model** menu (Settings → 2 Translate).



Which model?

- **claude-haiku-4-5** — the default. Fast (well under a second a line) and reliable.
- **gpt-6-sol** and **gpt-6.1-terra** (fast tier) — the best at repairing a misheard word from context. Good partners for the Decision Router.
- **claude-sonnet-5-5** — strong at context (who "sunt" or "am mers" refers to), about as fast as Sol. It runs with its thinking turned off, which stopped it putting its reasoning into captions.
- **none, fast** — Soniox's own translation, no model at all. The fastest and cheapest.

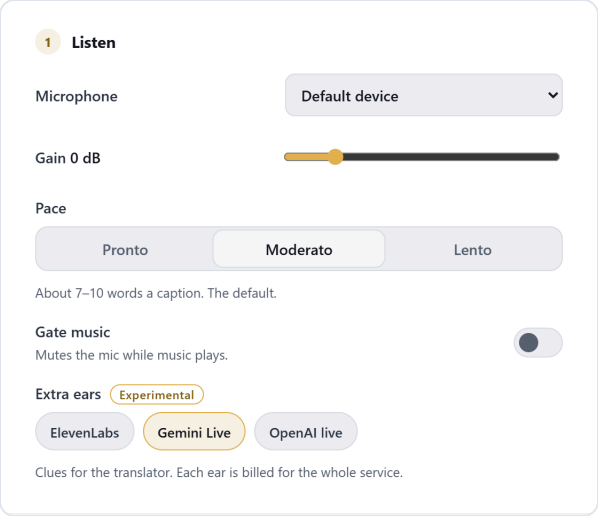
You can switch model in the middle of a service; the next line uses the new one. Models whose key is missing are greyed out. The model's light at the top turns green as it answers; its hollow second light shows whether the provider's prompt cache is saving you money.

6 Fewer mistakes: the Decision Router and extra ears

Both need an **OpenAI** key.

The Decision Router

Turn it on under **Settings** → **2 Translate**, right under **Model** (picture in Part 5). Soniox's quick translation of each phrase goes on screen straight away — the fastest captions Áron can give — but a small, fast AI (gpt-6-luna) checks every phrase first, in about a fifth of a second. When it doubts one — a clumsy or wrong translation, an unsuitable word, or a word that was probably misheard — that phrase goes to the model chosen in **Model** to be redone. On two recorded services it cut the lines not fit for the screen to about a third (from 9% to 3%, and from 5% to 2%). Its light at the top (**gpt-6-luna**) is dimmed while it is off. How doubtful a phrase must be is `CT_ROUTER_THRESHOLD` in `.env` (default 0.3; lower sends more lines on).



Extra ears

Speech-to-text sometimes mishears a word: a place name, a sound-alike. Extra ears (under **Settings** → **1 Listen**) are second and third recognisers — ElevenLabs, OpenAI live, Gemini Live — listening to the same audio. Their versions appear in grey under each line, and when the Decision Router suspects a mishearing, the translator is shown what each ear heard. On one sermon, 26 of 27 misheard words came out right with all three ears, against 20 with none. A doubtful phrase waits up to 0.7 s for them (`CT_EAR_WAIT_MS`).

> Each ticked ear listens to the whole service and is billed for it (about $0.40–1.00 an hour each). Start with **Gemini Live** alone: it was the most helpful single ear.

7 Fine-tuning

Pace — how much text is translated at a time

Stop	Words per piece	Good for
Pronto	about 3–7	captions as soon as possible
Moderato (default)	about 7–10	most services
Lento	about 10–14	the best translation: the model sees more context

Pace is under **Settings** → **1 Listen** (picture in Part 6). If captions break sentences in odd places, move it one step right; if they lag, one step left. Under the Decision Router, Soniox's own phrases set the size and Pace is greyed out.

Gate music

Switch on **Gate music** (Settings → 1 Listen) and Áron mutes itself while the band plays, so worship songs cost nothing and nothing odd reaches the screen. The **Mic** button (3) shows it: *Muted* in orange while the music plays, with a dotted underline while the music detector is in charge. Click it (or press `2` to mute, `1` to unmute) to take over by hand.

Every control and light explains itself when you hold the mouse over it. The readouts under the buttons (Pace, Voice, Wall) open Settings at the right place when clicked.

8 What it costs



The **$** at the end of the lights shows what this run has cost so far, across every service. Hold the mouse over it to see the split by service. It is an estimate at each provider's list price (stored in `config/prices.json`); your provider's own dashboard has the exact bill.

Set-up	Roughly, per hour of speech
Soniox only (speech-to-text + its own translation)	about $0.15
+ Soniox voice	+ $0.70
+ claude-haiku-4-5 translating every line	+ about $0.70
+ gpt-6-sol (fast) translating every line	+ about $2
Decision Router with gpt-6-sol redoing the doubtful lines	+ about $0.60
Each extra ear	+ $0.39 (ElevenLabs), $0.54 (Gemini Live), $1.02 (OpenAI live)

Estimates from list prices on 10 October 2026, assuming about 130 spoken words a minute. Prompt caching (re-using the unchanged part of each request) keeps the AI models cheap: a gpt-6-sol line costs about $0.002. Gate music and the idle stop keep you from paying for silence or songs.

9 Keeping the service

- The **download** button at the top right, beside the lights, offers two files. **Session record** (or `H`) downloads the session as a web page: original beside translation, what the extra ears heard, which model translated each line and why, and every setting with the time it changed. The same page is saved to the `recordings` folder as the service goes.
- With **Speak** on and **Record speech** on (Settings → 4 Voice), the spoken translation is recorded. **Spoken audio** (or `A`) downloads it as one mp3 with matching subtitles and a transcript. **Auto-save on exit** writes it to the `recordings` folder when you close Áron.

10 Troubleshooting and shortcuts

Problem	What to do
"SETUP REQUIRED: missing API key"	The `.env` file is missing, misnamed (Notepad may have saved it as `.env.txt`) or not beside the program (Part 1).
Nothing is transcribed	Check **Microphone** (Settings → 1 Listen), that the level bar moves, and that the **Mic** button is not on *Muted*. Make sure the page address is `localhost`: browsers only allow the microphone there.
"Soniox key not configured"	Open `.env` and check the `SONIOX_API_KEY=` line, then restart.
A model is greyed out	Its provider's key is missing from `.env` (Part 5), or it cannot write one of the languages you picked.
Speak does nothing	Check its light at the top. With only a Soniox key it speaks straight away. If it is silent, an **ElevenLabs** choice is selected under **Speech** without its key — switch **Speech** to **Soniox v2**, or add the ElevenLabs key.
Captions split sentences oddly / lag	Move **Pace** right / left (Part 7).
The Áron page was closed mid-service	Translation stops while it is closed; the black window keeps running. Open `http://localhost:5000` again and press **Start**. Within 20 minutes the transcript comes back (click **Start fresh** to begin anew instead); your settings, the recording and, within 5 minutes, the translator's thread carry on.
"Address already in use"	Áron is already running in another console window. Close it, or use that one.
Something else	A red light says where. Click **Diagnostics** (the pulse button at the top right) for the event log, and **Copy** it to whoever helps you.

Key	Does	Key	Does
`Space`	start / stop listening	`W`	clear the projector
`H`	download the transcript	`A`	download the spoken audio
`1` / `2`	Mic on / muted	`F11`	projector full screen

More detail: `QUICKSTART.md` (operator notes) and `README.md` (how it works) in the Áron folder.